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CONTROL SYSTEM, CONTROL METHOD, AND NON-TRANSITORY COMPUTER READABLE MEDIUM

Abstract

A control system includes a processor configured to: detect, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; identify, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; control an arm to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and remove the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation of International Application No. PCT/JP2023/029850 filed on Aug. 18, 2023, and claims priority from Japanese Patent Application No. 2023-025469 filed on Feb. 21, 2023.

BACKGROUND

(i) Technical Field

[0002] The present disclosure relates to a control system, a control method, and a non-transitory computer readable medium.

(ii) Related Art

[0003] Japanese Unexamined Patent Application Publication (Translation of PCT Application) No. 2017-519648 discloses a method for processing a stack of flexible substrates by using a gripper. The gripper is disposed on a robot arm and includes a lower finger and an upper finger corresponding to each other. The method includes a step of slipping the lower finger under a bottom-side substrate of the stack which is placed on a base and slipping the upper finger above a top-side substrate of the stack and a step of gripping the stack by tightening the stack between the bottom-side substrate and the top-side substrate. The method also includes: a step of raising a front part of the stack by the gripper using a first lift so that the entire stack is in a dangling state and raising the gripper until the stack faces vertically downward while the gripper is following the tilting state of the front part of the stack; a step of moving the stack in the dangling state to a support member by the gripper; a step of lowering the stack in the support member by the gripper so that a rear part of the stack at the opposite side of the front part of the stack contacts the support member; and a step of further lowering, before removing the gripper from the stack, the gripper while horizontally shifting the stack in a direction from the bottom-side substrate to the top-side substrate so as to arrange the entire stack on the support member in a state in which the bottom-side substrate faces upward.

SUMMARY

[0004] In humid regions, such as Asia, it is business customary to cover sheets of paper in wrapping paper and then to distribute them to reduce the influence of moisture on the sheets.

[0005] Before setting sheets in a printer, for example, it is thus necessary to remove a bundle of sheets from wrapping paper. A sheet of paper is more likely to bend than another type of sheet, such as a resin sheet. Additionally, when a user opens wrapping paper, he/she may damage sheets inside, and only a skilled operator does this work for taking out sheets of paper so far.

[0006] Aspects of non-limiting embodiments of the present disclosure relate to a control system, a control method, and a non-transitory computer readable medium that can make it possible to remove a bundle of sheets from wrapping paper without intervention of a skilled operator.

[0007] Aspects of certain non-limiting embodiments of the present disclosure address the above

advantages and/or other advantages not described above. However, aspects of the non-limiting embodiments are not required to address the advantages described above, and aspects of the non-limiting embodiments of the present disclosure may not address advantages described above.

[0008] According to an aspect of the present disclosure, there is provided a control system including a processor configured to: detect, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; identify, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; control an arm to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and remove the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] An exemplary embodiment of the present disclosure will be described in detail based on the following figures, wherein:

[0010] FIG. 1 illustrates an example of the configuration of a control system;

[0011] FIG. 2 illustrates an example of the inclination of a wrapper;

[0012] FIG. 3 illustrates an example of a gripper;

[0013] FIG. 4 illustrates an example of the functional configuration of a control device;

[0014] FIG. 5 illustrates a wrapper loaded on a loading table;

[0015] FIG. 6 illustrates examples of the major elements of the electrical system configuration of the control device;

[0016] FIG. 7 is a flowchart illustrating an example of sheet handling processing;

[0017] FIG. 8 illustrates that a wrapper is being shifted by an arm;

[0018] FIG. 9 illustrates an example of a movement path of knives;

[0019] FIG. 10 illustrates an example of the position at which a side surface of a wrapper is opened by a knife;

[0020] FIG. 11 illustrates an example of the state of a sheet bundle after an upper wrapper is removed;

[0021] FIG. 12 illustrates a state in which a sheet bundle is to be gripped by a gripper; and

[0022] FIG. 13 illustrates a state in which a sheet bundle is gripped by grippers.

DETAILED DESCRIPTION

[0023] An exemplary embodiment of the disclosure will be described below with reference to the accompanying drawings. The same element is designated by like reference numeral and the same operation is designated by like step number throughout the drawings and an explanation thereof will not be repeated.

(System Configuration)

[0024] FIG. 1 illustrates an example of the configuration of a control system 1 that handles sheets 14 used for printing, for example. The control system 1 includes arms 2 and a control device 10 that controls the arms 2. Each arm 2 is used for gripping an object with a gripper 8 attached to the forward end of the arm 2.

[0025] Sheets 14 used for printing are wrapped in wrapping paper 4 and are then transported to

reduce a quality change caused by moisture. The sheets **14** are thus placed in the form of a bundle of sheets **14** wrapped in the wrapping paper **4** on a loading table **3** installed within the operating ranges of the arms **2**. Specifically, the bundle of sheets **14** is placed on the loading table **3** so that the image forming surfaces of the sheets **14** oppose the loading surface of the loading table **3**. [0026] Hereinafter, the wrapping paper **4** wrapping a bundle of sheets **14** will be called a wrapper **4**. The shape of the sheets **14** to be handled in the control system **1** is not limited to a specific shape, and cut paper having a rectangular shape is used as the sheets **14** in one example in the exemplary embodiment. The shape of the wrapper **4** is accordingly a cuboid. To make it easy to place a bundle of sheets **14** wrapped in the wrapper **4**, the shape of the loading surface of the loading table **3** is also a rectangle. A bundle of sheets **14** is constituted by multiple sheets **14**, and hereinafter, it will be called a sheet bundle **14A**.

[0027] The control device **10** controls the arms **2** to open the wrapper **4** wrapping the sheet bundle **14A** placed on the loading table **3** with a knife **8B** (see FIG. 3) and to grip the sheet bundle **14A** exposed outside the wrapper **4** with the gripper **8** and to be removed from the wrapper **4**. The sheet bundle **14A** is then transported to a jogger **5** by the arms **2**. The knife **8B** is an example of a cutting device attached to the arm **2**.

[0028] The jogger **5** is a device that arranges the sheet bundle **14A** to align the outline of the individual sheets **14** before starting a working process, such as printing, collating, and binding, for example, using the sheets **14**.

[0029] A pillar **7** is mounted on a stand **16** on which the arms **2** are placed, and a 3D sensor **6** is attached to the forward end of the pillar **7**.

[0030] The 3D sensor **6** is an example of a three-dimensional sensor that measures the position, in a three-dimensional space, of the wrapper **4** that wraps the sheet bundle **14A**, which is placed on the loading table **3**.

[0031] The position of the wrapper **4** in the three-dimensional space is represented by the values of the three-dimensional coordinates set in the three-dimensional space (hereinafter simply called the three-dimensional coordinate values) and the inclination of the wrapper **4**.

[0032] In the control system **1**, a predetermined position in the three-dimensional space is set to the origin of the three-dimensional coordinates. In one example, one vertex on the loading surface of the loading table **3** is set to the origin of the three-dimensional coordinates in the exemplary embodiment. In this case, as seen from a position opposing the loading surface of the loading table **3**, the longitudinal direction of the loading table **3** is set to the X axis, the widthwise direction of the loading table **3** is set to the Y axis, and the vertical direction is set to the Z axis. The loading surface of the loading table **3** may not necessarily be horizontally placed with respect to the floor. For example, the loading surface of the loading table **3** may be tilted with respect to the floor so as to make it easy for the arms **2** to grip the wrapper **4**. In the exemplary embodiment, however, it is assumed that the loading surface of the loading table **3** is horizontally placed with respect to the floor.

[0033] The 3D sensor **6** measures the distances to the individual points of the wrapper **4** to detect the three-dimensional coordinate values of the wrapper **4** at the individual points. Detecting the three-dimensional coordinate values of the wrapper **4** at the individual points also means detecting the shape of the wrapper **4**. As the 3D sensor **6**, various sensors that can measure the distance to an object is used. Examples of such sensors are a LiDAR (Light Detection And Ranging) sensor, a ToF (Time of Flight) sensor, and a stereo camera.

[0034] The 3D sensor **6** also detects the inclination of the wrapper **4** by using the detected three-dimensional coordinate values of the wrapper **4**. The inclination of the wrapper **4** is the inclination of the wrapper **4** with respect to a predetermined reference line in the three-dimensional space.

[0035] If the X axis is set to the reference line, for example, the inclination of the wrapper **4** is represented by the angle θ formed between the X axis in the XY plane and the long side of the wrapper **4** loaded on the loading table **3**. That is, the inclination of the wrapper **4** also serves as an

index indicating the state of the wrapper **4** placed on the loading table **3**, such as how exact the wrapper **4** is parallel with the sides of the loading table **3**.

[0036] The above-described representation of the inclination of the wrapper **4** is only an example. The inclination of the wrapper **4** may be represented in a different manner. For example, if, as shown in FIG. **2**, a different wrapper **4** is placed adjacent to and under a wrapper **4** whose inclination is to be detected, the long side of the different wrapper **4** in the XY plane may be set to a reference line, and the angle θ formed between the long sides of the two wrappers **4** may be used as the inclination of the wrapper **4**.

[0037] In the example in FIG. **1**, the 3D sensor **6** is attached to the pillar **7**. However, the 3D sensor **6** may be placed at another location if it is a location where the 3D sensor **6** can see the entirety of the wrapper **4** in the Z-axis direction, such as above the image forming surfaces of the sheets **14** wrapped in the wrapper **4**, namely, above the wrapper **4**, and the relative position with the wrapper **4** is not changed. In one example, the 3D sensor **6** may be fixed to the ceiling of a room where the arms **2** are installed. The 3D sensor **6** may be attached to the arm **2**. In this case, however, to detect the three-dimensional coordinate values and the inclination of the wrapper **4**, it is necessary to move the 3D sensor **6** along the shape of the wrapper **4** or to move the 3D sensor **6** to a position at which it can see the entirety of the wrapper **4** and then to cause it to detect the three-dimensional coordinate values and the inclination of the wrapper **4**.

[0038] Additionally, since the arm **2** is moving, it is also necessary to consider the position of the arm **2** to detect the three-dimensional coordinate values and the inclination of the wrapper **4**. Because of the above-described reasons, using the 3D sensor **6** attached to the arm **2** may take more time to detect the three-dimensional coordinate values and the inclination of the wrapper **4** than using the 3D sensor **6** mounted on a fixed location, such as the pillar **7** or the ceiling of the room.

[0039] Although the control system **1** includes two arms **2** in the example in FIG. **1**, there is no limitation on the number of arms **2**. One arm **2** or three or more arms **2** may be provided in the control system **1**. Nevertheless, if the control system **1** includes only one arm **2**, the sheet bundle **14A** is gripped by one gripper **8**, which makes it more difficult to move the sheet bundle **14A** than when multiple grippers **8** grip the sheet bundle **14A**. Hence, the use of multiple arms **2** may be desirable. On the other hand, however, as more arms **2** are provided, the cost of the control system **1** becomes higher, and the control operation also becomes complicated. Thus, it may be desirable to provide two arms **2** in the control system **1**.

(Structure of Gripper)

[0040] FIG. **3** illustrates an example of the gripper **8** attached to the forward end of the arm **2**. The gripper **8** includes two claws **9**, a knife **8B**, and a suction portion **8A**. The claws **9** grip an object under the control of the control device **10**, for example. The knife **8B** is used for opening the wrapper **4**. The suction portion **8A** sucks pieces of the opened wrapper **4**.

[0041] The claws **9** are constituted by a pair of claws **9A** and **9B**. The claw **9A** is a circular claw whose forward end is formed in a circle. An elastic member, such as rubber, is fixed to the contact surface of the claw **9A**, which is to contact the sheet bundle **14A**, to make it easy to grip the sheet bundle **14A**. The claw **9B** is formed in the shape of a spatula, and the contact surface with the sheet bundle **14A** is formed in a planar shape to make it easy to grip the sheet bundle **14A** from downward. The claw **9B**, which is formed in a spatula-like shape, can be easily slipped into, for example, a gap between two sheets **14** or a gap between the opened wrapper **4** and the sheet bundle **14A**, compared with the claw **9A**. An elastic member is also fixed to the contact surface of the claw **9B** to contact the sheet bundle **14A**. Hereinafter, the claws **9A** and **9B** will be called the claw **9** if it is not necessary to individually distinguish them from each other.

[0042] The knife **8B** also serves the following function, for example, as well as for opening the wrapper **4**. The knife **8B** is slipped into a gap between the opened wrapper **4** and the sheet bundle **14A** and raises the sheet bundle **14A** to form a space therebetween, thereby making it easy for the claw **9B** to be inserted between the wrapper **4** and the sheet bundle **14A**.

[0043] The suction portion **8A** sucks the pieces of the wrapper **4** through a suction port by utilizing air suction force. The suction structure of the suction portion **8A** is not limited to a particular structure. For example, the suction portion **8A** may use a gel-like sheet having adhesiveness to suck the pieces of the wrapper **4**. The gripper **8** may not necessarily include the suction portion **8A**. In this case, the gripper **8** holds a member corresponding to the suction portion **8A** with the claw **9** to suck the pieces of the wrapper **4**.

(Functional Configuration of Control Device)

[0044] FIG. **4** illustrates an example of the functional configuration of the control device **10**. The control device **10** includes functional elements, such as a position detector **11**, a side surface identifier **12**, and a controller **13**.

[0045] By using the 3D sensor **6**, the position detector **11** detects the three-dimensional coordinate values and the inclination representing the position and the shape of the wrapper **4** and those of the sheet bundle **14A** wrapped in the wrapper **4**.

[0046] The side surface identifier **12** identifies the locations of side surfaces **4S** of the wrapper **4** by using the three-dimensional coordinate values and the inclination detected by the position detector **11**. Among the surfaces of the cuboid wrapper **4**, the side surfaces **4S** are surfaces intersecting with the image forming surface of each sheet **14** wrapped in the wrapper **4**.

[0047] FIG. **5** illustrates the wrapper **4** loaded on the loading table **3**. In FIG. **5**, the surfaces along the Z-axis direction indicated by the hatched portions, that is, the surfaces of the wrapper **4** in the XZ plane and the YZ plane, are the side surfaces **4S** of the wrapper **4**. Among the surfaces opposing the image forming surfaces of the sheets **14** wrapped in each wrapper **4**, that is, among the surfaces of the wrapper **4** in the XY plane, the upper surface will be called a top surface **4T** of the wrapper **4**, while the lower surface will be called a bottom surface **4U** of the wrapper **4**.

[0048] The controller **13** controls each arm **2** so that the knife **8B** of the gripper **8** can move on the side surfaces **4S** identified by the side surface identifier **12** along the outlines of the sheets **14** wrapped in the wrapper **4** so as to open the wrapper **4**. The controller **13** also controls each arm **2** to grip the sheet bundle **14A** exposed outside the wrapper **4** whose side surfaces **4S** are opened by the knife **8B** of the gripper **8** and to take out the sheet bundle **14A** from the wrapper **4**.

(Configuration of Control Device Using Computer)

[0049] The control device **10** executing the above-described processing is constituted by a computer **20**. FIG. **6** illustrates examples of the major elements of the electrical system configuration of the control system **10** constituted by the computer **20**.

[0050] The computer **20** includes a central processing unit (CPU) **21**, a random access memory (RAM) **22**, a non-volatile memory **23**, and an input/output interface (I/O) **24**. The CPU **21** is an example of a processor that executes processing operations of the individual functional elements of the control device **10** shown in FIG. **4**. The RAM **22** is used as a temporary work area for the CPU **21**. The CPU **21**, RAM **22**, non-volatile memory **23**, and I/O **24** are connected to one another via a bus **25**.

[0051] The non-volatile memory **23** is an example of a storage device that can retrieve stored information even after power supplied to the non-volatile memory **23** is interrupted. As the non-volatile memory **23**, a semiconductor memory, for example, is used, or a hard disk may alternatively be used. In the non-volatile memory **23**, a control program that allows the computer **20** to function as the control device **10**, for example, is stored.

[0052] The non-volatile memory **23** may not necessarily be built in the computer **20**, and may be a portable storage device that is attachable to and detachable from the computer **20**, such as a memory card.

[0053] The 3D sensor **6** and the arms **2** are connected to the I/O **24**. Devices connected to the I/O **24** are not limited to these devices. For example, another device, such as a communication device that communicates with an external device via a communication network, such as the internet or a local area network (LAN), may be connected to the I/O **24**.

(Operation of Control System)

[0054] The operation of the control system **1** for taking out the sheet bundle **14A** from the wrapper **4** by using the arms **2** will now be explained below.

[0055] FIG. **7** is a flowchart illustrating an example of sheet handling processing executed by the CPU **21** of the control device **10** in response to a user instruction to take out the sheet bundle **14A** from the wrapper **4**.

[0056] The control program describing the sheet handling processing is prestored in the non-volatile memory **23** of the control device **10**, for example. The CPU **21** of the control device **10** reads the control program from the non-volatile memory **23** and executes the sheet handling processing. It is assumed that multiple sheet bundles **14A** which are each wrapped in a wrapper **4** are placed on the loading table **3**.

[0057] In step **S10**, the CPU **21** controls the 3D sensor **6** to detect the three-dimensional coordinate values and the inclination of a wrapper **4** wrapping the sheet bundle **14A** placed at the topmost position on the loading table **3**. The wrapper **4** described below is the wrapper **4** wrapping the sheet bundle **14A** placed at the topmost position on the loading table **3**. The wrapper **4** wrapping the sheet bundle **14A** placed at the topmost position on the loading table **3** is an example of a wrapper **4** to be opened under the control of the control system **1**. The three-dimensional coordinate values and the inclination of the wrapper **4** will be called work information.

[0058] In step **S20**, by using the work information obtained in step **S10**, the CPU **21** recognizes the position and the shape of the wrapper **4** and identifies the locations of the side surfaces **4S** of the wrapper **4**.

[0059] In step **S30**, the CPU **21** controls an arm **2** so that the claw **9** of the gripper **8** contacts a side surface **4S** of the wrapper **4** and pushes out the wrapper **4** horizontally in a direction intersecting with the direction along the side surface **4S** of the wrapper **4**, that is, the claw **9** pushes out the wrapper **4** along the XY plane. The side surface **4S** of the wrapper **4** that the claw **9** contacts may be either one of the side surface including the long side of the wrapper **4** and the side surface **4S** including the short side of the wrapper **4**.

[0060] The CPU **21** may control multiple arms **2** to push out the wrapper **4**. However, pushing out the wrapper **4** with one arm **2** can reduce the control time for the arm **2** to shift the wrapper **4**.

[0061] With the above-described control operation, one end of the wrapper **4** is displaced from that of another wrapper **4** underneath in the X-axis direction or the Y-axis direction.

[0062] FIG. **8** illustrates that the wrapper **4** is being shifted by the arm **2**. In the example in FIG. **8**, the arm **2** pushes out the side surface **4S** including the short side of the wrapper **4** to shift the wrapper **4** in the X-axis direction. Shifting the position of the wrapper **4** from another wrapper **4** underneath in this manner can form a space at a front portion on the bottom surface **4U** of the wrapper **4** for the claw **9B** to slip into, thereby making it easy for the claw **9** of the gripper **8** to grip the sheet bundle **14A** wrapped in the wrapper **4**.

[0063] When pushing out the wrapper **4** under the control of the CPU **21**, if the displacement amount from the side surface **4S** of the wrapper **4** to that of another wrapper **4** underneath in the shifting direction, that is, the moving distance **D** of the wrapper **4**, is too short, it makes it difficult for the claw **9** of the gripper **8** to grip the sheet bundle **14A**. There is a prescribed distance that allows the claw **9** of the gripper **8** to easily grip the sheet bundle **14A**. It may be thus desirable that the CPU **21** control the arm **2** to push out the wrapper **4** with a certain strength of force that can shift the wrapper **4** by the amount of the moving distance **D** equal to or longer than this prescribed distance.

[0064] To implement this, the CPU **21** obtains attribute values representing attributes of the sheets **14** wrapped in the wrapper **4** and sets the strength of force used by the arm **2** to push out the wrapper **4**. An attribute of the sheets **14** is a characteristic of the sheets **14**, and the attribute value is a value representing this characteristic of the sheets **14**.

[0065] The sheets **14** have various attributes, such as the number of sheets **14** forming the sheet

bundle **14A** wrapped in the wrapper **4** (the storage number of sheets in the wrapper **4**), and the size (A1 or A2, for example), and ream weight, basis weight, thickness, and moisture content of the sheets **14**. The strength of force for pushing out the wrapper **4** is subject to the weight of the sheet bundle **14A** wrapped in the wrapper **4**. The CPU **21** thus obtains attribute values required for calculating the weight of the sheet bundle **14A** wrapped in the wrapper **4**. More specifically, the CPU **21** obtains the ream weight or the basis weight of the sheets **14**, the size of the sheets **14**, and the storage number of sheets **14** in the wrapper **4** and calculates the weight of the sheet bundle **14A** wrapped in the wrapper **4**.

[0066] The strength of force for pushing out the wrapper **4** is also subject to the friction coefficient on the surface of the wrapper **4**. The CPU **21** thus obtains information on the material of the wrapper **4**, such as whether the wrapper **4** is kraft paper or coated paper and, if it is kraft paper, whether it is bleached kraft paper or unbleached kraft paper, and then sets the strength of force of the arm **2** to push out the wrapper **4** by taking the friction coefficient determined from the material of the wrapper **4** into consideration. The friction coefficient of recycled paper and that of non-recycled paper may be different from each other. The CPU **21** may thus obtain information indicating whether the wrapper **4** is recycled paper as an attribute value of the wrapper **4**.

[0067] A user prestores the attribute values of the wrapper **4** in the non-volatile memory **23** before causing the CPU **21** to execute the sheet handling processing. The CPU **21** then obtains the attribute values from the non-volatile memory **23**. However, the CPU **21** may obtain the attribute values of the wrapper **4** by a different approach. For example, a label indicating the attribute values of the wrapper **4** may be fixed to the wrapper **4**. In this case, the CPU **21** may perform control, for example, so that the claw **9** of the gripper **8** holds a visible camera (not shown) and the visible camera takes an image of the attribute values of the wrapper **4** indicated on the label. Then, the CPU **21** performs known character recognition processing on the captured image to obtain the attribute values of the wrapper **4**. The visible camera may be fixed to the gripper **8** in advance.

[0068] If the attribute values of the wrapper **4** indicated on the label are represented by a barcode, the CPU **21** may perform control so that the claw **9** of the gripper **8** holds a barcode reader (not shown) and the barcode reader reads the barcode to obtain the attribute values of the wrapper **4**. The barcode reader may be fixed to the gripper **8** in advance.

[0069] In step **S30** in FIG. 7, the CPU **21** controls the arm **2** to push out the wrapper **4** horizontally along the XY plane by way of example. It is sufficient, however, if the wrapper **4** is displaced from another wrapper **4** underneath. For example, the CPU **21** may control the arm **2** to pull the wrapper **4** horizontally along the XY plane while the wrapper **4** is being held by the claw **9** of the gripper **8**.

[0070] Since the position of the wrapper **4** is displaced in step **S30**, the work information of the wrapper **4** becomes different from that obtained in step **S10**. In step **S40**, the CPU **21** controls the 3D sensor **6** to obtain the work information of the wrapper **4** and to re-recognize the position and the shape of the wrapper **4**, thereby identifying the locations of the side surfaces **4S** of the displaced wrapper **4**.

[0071] In step **S50**, the CPU **21** determines whether the moving distance **D** of the displaced wrapper **4** is larger than or equal to the prescribed distance. If the moving distance **D** of the wrapper **4** is smaller than the prescribed distance, the CPU **21** returns to step **S20** and repeats steps **S20** through **S50**. That is, the CPU **21** re-identifies the locations of the side surfaces **4S** of the displaced wrapper **4** and executes steps **S30** and **40** until it is determined in step **S50** that the moving distance **D** of the displaced wrapper **4** is larger than or equal to the prescribed distance.

[0072] If it is determined in step **S50** that the moving distance **D** of the displaced wrapper **4** is larger than or equal to the prescribed distance, the CPU **21** proceeds to step **S60**.

[0073] In step **S60**, the CPU **21** controls the arm **2** so that the knife **8B** of the gripper **8** is moved along the outlines of the sheets **14** while keeping a position at which the knife **8B** faces the side surfaces **4S** of the wrapper **4** so as to open the side surfaces **4S**.

[0074] The wrapper **4** may be of a type that is bent and is difficult to open when an edged tool is

pressed against the wrapper **4**. To handle such a wrapper **4**, an ultrasonic cutter may be used as the knife **8B** of the gripper **8** so as to make it easy to open the wrapper **4**. However, a regular knife, which is not provided with a vibrator producing vibrations, unlike an ultrasonic cutter, may be used.

[0075] FIG. **9** illustrates an example of the movement path of the knives **8B** of the grippers **8** of the arms **2** moved under the control of the CPU **21** when the top surface **4T** of the wrapper **4** is seen from above in the Z-axis direction. As illustrated in FIG. **9**, the CPU **21** controls the arms **2** so that the knives **8B** of the grippers **8** are moved to positions of the side surfaces **4S** at which the knives **8B** face each other across the wrapper **4**. The CPU **21** then controls the arms **2** so that the knives **8B** can be kept contacting the corresponding side surfaces **4S** of the wrapper **4** and be each moved clockwise or counterclockwise by half a turn along the outlines of the sheets **14** while maintaining the positions at which the knives **8B** face the corresponding side surfaces **4S**. With this operation, the wrapper **4** wrapping the sheet bundle **14A** is separated into two upper and lower pieces.

[0076] It is sufficient if the wrapper **4** is opened to expose the sheet bundle **14A** to such a degree as to allow the claws **9** of the grippers **8** to grip the sheet bundle **14A**. From this viewpoint, the wrapper **4** may not necessarily be opened all around the side surfaces **4S**, that is, all around the four sides of the wrapper **4**, and be separated into two upper and lower pieces. For example, the side surfaces **4S** of the wrapper **4** may be opened along the outlines of three sides of the wrapper **4**. If, however, the side surfaces **4S** of the wrapper **4** are opened along the outlines of two or less sides of the wrapper **4**, for example, when the grippers **8** gripping the sheet bundle **14A** are moved upward in the Z-axis direction, the sheet bundle **14A** may be caught by the wrapper **4** and may not be able to be removed from the wrapper **4**. It may be thus desirable that the side surfaces **4S** of the wrapper **4** be opened along the outlines of three or more sides of the wrapper **4** with the knives **8B** of the grippers **8**.

[0077] The side surfaces **4S** of the wrapper **4** may be opened by moving the knife **8B** of the gripper **8** of one arm **2** or the knives **8B** of the grippers **8** of three or more arms **2** along the outlines of the sheets **14** under the control of the CPU **21**. In the case of the use of one arm **2**, the CPU **21** controls the arm **2** so that the knife **8B** of the gripper **8** is moved by one turn around the side surfaces **4S** of the wrapper **4** along the outlines of the sheets **14**. In the case of the use of three arms **2**, for example, the CPU **21** controls the arms **2** so that the knives **8B** of the grippers **8** of the arms **2** are each moved by $\frac{1}{3}$ turns of the side surfaces **4S** of the wrapper **4** along the outlines of the sheets **14** without the movement paths of the knives **8B** overlapping each other.

[0078] Instead of the knife **8B** of the gripper **8**, another cutting device (not shown), such as a cutter, may be used to open the side surfaces **4S** of the wrapper **4** while being carried by the claw **9** of the gripper **8**. The cutting device carried by the claw **9** of the gripper **8** is an example of a cutting device attached to the arm **2**.

[0079] When opening the side surfaces **4S** of the wrapper **4** as described above, the CPU **21** controls the arm **2** so that the knife **8B** of the gripper **8** opens the side surface **4S** at a position included in a range above the center of the side surface **4S** and below the top edge of the side surface **4S**.

[0080] FIG. **10** illustrates an example of the position at which the side surface **4S** of the wrapper **4** is opened by the knife **8B** of the gripper **8**. The side surface **4S** of the wrapper **4** shown in FIG. **10** is a side surface **4S** in the XZ plane seen from a position that opposes the side surface **4S** in the Y-axis direction.

[0081] It is assumed that, for example, the side surface **4S** of the wrapper **4** is opened at a position on a line **L1** which represents the bottom edge of the side surface **4S** of the wrapper **4**. In this case, to grip the sheet bundle **14A** by the gripper **8**, one of the claws **9**, and more specifically, the spatula-like claw **9B**, which is easier to be inserted than the claw **9A**, is slipped into a gap between the sheet **14** located at the bottommost position of the sheet bundle **14A** and a piece of the wrapper **4** underneath.

[0082] To insert the claw **9B** into this gap, however, it is necessary that the CPU **21** control the position of the claw **9B** with a precision of micrometers or smaller. Even if the CPU **21** successfully controls the position of the claw **9B** with such a high precision, it is still difficult to insert the claw **9B** having a certain thickness into the above-described gap.

[0083] Accordingly, the side surface **4S** is opened at a position at least above the line **L1**, thereby allowing a paper scrap forming the side surface **4S** of the wrapper **4** (hereinafter simply called a scrap) to remain in the lower piece of the wrapper **4**. With a scrap remaining in the lower piece of the wrapper **4**, the claw **9B** can be guided to farther inward than the scrap while holding down the scrap from above and be inserted into the gap between the sheet **14** located at the bottommost position of the sheet bundle **14A** and the lower piece of the wrapper **4** underneath by using the scrap as a support.

[0084] Nevertheless, if the width of the scrap remaining in the lower piece of the wrapper **4** is short, it becomes difficult for the claw **9B** to hold down the scrap. Hence, a portion above a line **L2** which represents the center of the side surface **4S** of the wrapper **4** shown in FIG. **10** is opened so that a scrap whose width is larger than half the width of the side surface **4S** of the wrapper **4** in the Z-axis direction can remain in the lower piece of the wrapper **4**. This makes it easier for the claw **9B** to hold down the scrap and to be guided to farther inward than the scrap, compared with when the side surface **4S** of the wrapper **4** is opened at a position in a range from the line **L1** to the line **L2**.

[0085] On the other hand, however, if the width of the scrap remaining in the lower piece of the wrapper **4** is too long, when the claw **9B** holds down the scrap, the sheet bundle **14A** may not be exposed until the position of the line **L1** due to the tension of the wrapper **4**. This makes it difficult to insert the claw **9B** into the gap between the sheet **14** located at the bottommost position of the sheet bundle **14A** and the lower piece of the wrapper **4** underneath.

[0086] For the above-described reason, the side surface **4S** of the wrapper **4** may be opened at a position in a range above the line **L2** representing the center of the side surface **4S** until a line **L3**, which is preset at a position below a line **LA** representing the top edge of the side surface **4S**.

[0087] For the sake of description, the lower piece of the wrapper **4** generated by opening the side surface **4S** in step **S60** will be called a lower wrapper **4A**, while the upper piece of the wrapper **4** generated by opening the side surface **4S** will be called an upper wrapper **4B**.

[0088] After the side surface **4S** of the wrapper **4** is opened, in step **S70** in FIG. **7**, the CPU **21** controls the arms **2** so that the suction portions **8A** suck and remove the upper wrapper **4B**, which forms the upper and lower pieces of the wrapper **4**.

[0089] FIG. **11** illustrates an example of the state of the sheet bundle **14A** when the side surface **4S** of the wrapper **4** and a side surface **14S** of the sheet bundle **14A** in the XZ plane are seen from a position that opposes the side surfaces **4S** and **14S** in the Y-axis direction after the upper wrapper **4B** is removed. As shown in FIG. **11**, the upper side of the sheet bundle **14A** is exposed, while the lower wrapper **4A** remains on the lower side of the sheet bundle **14A**.

[0090] The side surface **4S** of the wrapper **4** may not be able to be completely opened in step **S60**, and part of the upper wrapper **4B** may remain connected to the lower wrapper **4A**. In this case, after the upper wrapper **4B** is sucked by the suction portion **8A**, while the arm **2** is being moved to discard the upper wrapper **4B** in a trash can, the upper wrapper **4B** may come off from the suction portion **8A** and may not be able to be removed.

[0091] Hence, in step **S80**, the CPU **21** determines whether the removal of the upper wrapper **4B** has been completed. When the suction portion **8A** is sucking the upper wrapper **4B** with suction force, a load is applied to a motor generating the suction force, unlike when the suction portion **8A** is not in sucking operation. It can thus be determined that the removal of the upper wrapper **4B** has been completed if the above-described load is continuously applied to the motor during the period from the start of sucking the upper wrapper **4B** using the suction portion **8A** until the disposal of the upper wrapper **4B** in a trash can. The CPU **21** may determine whether the removal of the upper

wrapper 4B has been completed by using the 3D sensor 6.

[0092] If the removal of the upper wrapper 4B has not been completed, the CPU 21 obtains work information of the wrapper 4 by using the 3D sensor 6 and returns to step S60. In this case, the CPU 21 repeats steps S60 through S80 until it is determined in step S80 that the removal of the upper wrapper 4B has been completed. That is, the CPU 21 controls the arms 2 to reopen a portion of the side surface 4S of the wrapper 4 that has not been opened.

[0093] If it is determined in step S80 that the removal of the upper wrapper 4B has been completed, the CPU 21 proceeds to step S90.

[0094] While the upper wrapper 4B is being removed, it may contact the sheet bundle 14A that has been wrapped in the wrapper 4, and one or more sheets forming the sheet bundle 14A may be displaced from the other sheets 14 horizontally along the XY plane. If a sheet 14 is displaced from the other sheets 14 horizontally, it may become more difficult for the claw 9 of the gripper 8 to grip the sheet bundle 14A than when the side surface 14S of the sheet bundle 14A is flat.

[0095] Hence, in step S90, the CPU 21 controls the arm 2 so that the spatula-like claw 9B of the gripper 8 contacts the side surface 14S of the sheet bundle 14A and make the side surface 14S flat. As a result, the side surfaces of the sheets 14 forming the sheet bundle 14A become flush with each other. "Making the side surface 14S of the sheet bundle 14A flat" refers to a state in which, when the sheet bundle 14A is seen from a position above the image forming surfaces of the exposed sheets 14 in the Z-axis direction, the positions of the outlines of the individual sheets 14 forming the sheet bundle 14A are contained within a predetermined range.

[0096] In step S100, the CPU 21 controls the 3D sensor 6 to obtain work information of the sheet bundle 14A after the side surface 14S is made flat.

[0097] In step S110, the CPU 21 recognizes the position and the shape of the sheet bundle 14A by using the work information of the sheet bundle 14A obtained in step S100, and controls the arm 2 so that the claws 9 of the grippers 8 can grip the sheet bundle 14A exposed outside the wrapper 4. At this time, the CPU 21 determines the position of the sheet bundle 14A to be gripped by the claws 9 of the grippers 8 in accordance with the type of work. In this case, the claw 9 may grip a corner of the sheet bundle 14A by considering the balance to move the sheet bundle 14A. However, if it is necessary to turn over the sheet bundle 14A during working, the claw 9 may grip a portion near the center of a side of the sheet bundle 14A.

[0098] FIG. 12 illustrates a state in which the sheet bundle 14A is to be gripped by the claw 9 of the gripper 8. As discussed above, FIG. 12 illustrates a state in which the claw 9B is being guided to farther inward than a scrap of the lower wrapper 4A while holding down the scrap from above.

[0099] After the sheet bundle 14A is gripped by one arm 2, the CPU 21 controls the arm 2 to raise the gripped sheet bundle 14A, as shown in FIG. 12. The CPU 21 then controls the other arm 2 so that the claw 9B of this arm 2 is located under the claw 9A in a state in which the distance between the claws 9A and 9B is wider than the thickness of the sheet bundle 14A and that the claw 9B is inserted into the gap between the sheet bundle 14A and the lower wrapper 4A. While this state is being maintained, the CPU 21 controls this arm 2 so that the claws 9A and 9B are moved along the outlines of the sheets 14 and grip the corner of the sheet bundle 14A diagonally positioned on the opposite side of the corner of the sheet bundle 14A that has already been gripped by the other arm 2. FIG. 13 illustrates an example in which the sheet bundle 14A is gripped by the grippers 8.

[0100] In step S120 in FIG. 7, the CPU 21 determines whether the claws 9 of the grippers 8 of the individual arms 2 have gripped the sheet bundle 14A. If the claws 9 of the grippers 8 have not yet gripped the sheet bundle 14A, the CPU 21 obtains work information of the sheet bundle 14A by using the 3D sensor 6 and returns to step S90. The CPU 21 repeats steps S90 through S120 until it is determined in step S120 that the claws 9 of the grippers 8 have gripped the sheet bundle 14A. That is, the CPU 21 controls the arms 2 so that the side surface 14S of the sheet bundle 14A can be made flat and be gripped.

[0101] If it is determined in step S120 that the claws 9 of the grippers 8 have gripped the sheet

bundle **14A**, the CPU **21** proceeds to step **S130**.

[0102] The sheets **14** forming the sheet bundle **14A** may stick to their adjacent sheets **14** due to static electricity or friction of the sheets **14**.

[0103] In step **S130**, the CPU **21** performs control to execute wind supply to supply wind to the sheet bundle **14A** from a vent (not shown) provided in the gripper **8** so as to make it easy to separate the sheets **14** from each other one by one. In wind supply, using ionic wind rather than air makes it easier to remove static electricity occurring in the sheets **14** so as to facilitate the separation of the sheets **14**.

[0104] In step **S140**, the CPU **21** controls the arms **2** so that the sheet bundle **14A** gripped by the claws **9** of the grippers **8** is moved to the jogger **5**.

[0105] In step **S150**, the CPU **21** controls the arms **2** so that the suction portions **8A** suck and remove the lower wrapper **4A** remaining after the sheet bundle **14A** is taken out of the wrapper **4**.

[0106] In step **S160**, the CPU **21** determines whether the removal of the lower wrapper **4A** has been completed. If the lower wrapper **4A** has not been removed, the CPU **21** obtains work information of the lower wrapper **4A** by using the 3D sensor **6** and returns to step **S150**. In this case, the CPU **21** repeats steps **S150** and **S160** until it is determined in step **S160** that the removal of the lower wrapper **4A** has been completed. That is, the CPU **21** controls the arms **2** to remove the lower wrapper **4A**.

[0107] If it is determined in step **S160** that the removal of the lower wrapper **4A** has been completed, it means that the sheet bundle **14A** has been taken out of the wrapper **4**, and the CPU **21** finishes the sheet handling processing shown in FIG. **7**.

[0108] The CPU **21** executes the sheet handling processing shown in FIG. **7** for each wrapper **4** placed on the loading table **3** so as to take out the sheet bundle **14A** from each wrapper **4** and to move the sheet bundle **14A** to the jogger **5**. The CPU **21** then starts the subsequent work process using the sheets **14**.

[0109] An example of the control system **1** that handles the sheets **14** has been discussed through illustration of the exemplary embodiment. However, the disclosed mode of the control system **1** is only an example and is not limited to that in the exemplary embodiment. Various modifications and/or improvements may be made to the exemplary embodiment without departing from the spirit and scope of the disclosure. Exemplary embodiments obtained by making modifications and/or improvements are also encompassed within the technical range of the disclosure.

[0110] For example, the order of steps of sheet handling processing illustrated in FIG. **7** may be changed without departing from the spirit and scope of the disclosure.

[0111] In the above-described exemplary embodiment, sheet handling processing is implemented by software by way of example. However, processing equivalent to sheet handling processing in the flowchart in FIG. **7** may be implemented by using hardware. In this case, sheet handling processing is executed faster than using software.

[0112] In the above-described exemplary embodiment, the control program is stored in the non-volatile memory **23** by way of example. However, the control program may be stored in a location other than the non-volatile memory **23**. The control program of the disclosure may be recorded in a storage medium that can be read by the computer **20** and be provided.

[0113] For example, the control program may be provided as a result of being recorded in an optical disc, such as a compact disc read only memory (CD-ROM), a digital versatile disc read only memory (DVD-ROM), or a Blu-ray disc. The control program may be provided as a result of being recorded in a portable semiconductor memory, such as a universal serial bus (USB) memory or a memory card. The non-volatile memory **23**, a CD-ROM, a DVD-ROM, a Blu-ray disc, a USB memory, and a memory card are examples of a non-transitory storage medium.

[0114] Additionally, the control system **1** may download the control program from an external device connected to a communication network via a communication unit and store the downloaded control program in the non-volatile memory **23** of the control device **10**. In this case, the CPU **21** of

the control device **10** reads this downloaded control program from the non-volatile memory **23** and executes sheet handling processing.

[0115] In the exemplary embodiments, the processes are performed by any computer. The computer may perform the processes by using a processor serving as hardware, a program serving as software, or combination of these. In this case, the processor is configured to perform the processes in the exemplary embodiments in cooperation with the program and may function as a unit or a means in the exemplary embodiments. The order in which the processor performs the processes is not limited to the described order and may be changed appropriately. The computer may be a general-purpose computer, an application specific computer, a workstation, or another system capable of performing the processes.

[0116] The processor may be composed of one or more pieces of hardware, and the type of the hardware is not limited. For example, the processor may be composed of hardware such as a central processing unit (CPU **21**, for example), a micro processing unit (MPU), a programmable logic device such as a field programmable gate array (FPGA), a dedicated circuit for performing specific processing such as an application specific integrated circuit (ASIC), a graphics processing unit (GPU), or a neural processing unit (NPU). Regarding the type of the hardware, different types of hardware may be combined. If multiple pieces of hardware are configured to perform one or more processes of the processor, the multiple pieces of hardware may be present in apparatuses physically away from each other or may be present in one apparatus. In each of exemplary embodiments, the order in which the processor performs the processes is not limited to the order described above and may be changed appropriately. The hardware is composed of electric circuitry in which circuit elements such as semiconductor devices are combined, or the like.

[0117] Further, the program may be software such as firmware or microcode. The program may be, for example, a program module group, and the functions thereof may be implemented by processors configured to implement the respective functions. The program may be program code or multiple code segments stored in one or more non-transitory computer readable media (for example, a storage medium or another storage). The program may be stored in such a divided manner in multiple non-transitory computer readable media present in apparatuses physically away from each other. The program code or the code segments may represent a procedure, a function, a sub program, a routine, a subroutine, a module, a software package, a class or any combination of instructions, data structures, or program statements. The program code or the code segment may be connected to another code segment or a hardware circuit by transmitting and/or receiving information, data, an argument, a parameter, or memory content.

[0118] The foregoing description of the exemplary embodiments of the present disclosure has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Obviously, many modifications and variations will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, thereby enabling others skilled in the art to understand the disclosure for various embodiments and with the various modifications as are suited to the particular use contemplated. It is intended that the scope of the disclosure be defined by the following claims and their equivalents.

APPENDIX

((((1)))

[0119] A control system comprising: [0120] a processor configured to: [0121] detect, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; [0122] identify, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a

surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; [0123] control an arm to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and [0124] remove the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.

((2)))

[0125] The control system according to ((1))), wherein the processor is configured to: [0126] control the arm, before opening the side surface of the wrapping paper, so that the arm contacts the side surface to shift the wrapping paper in a direction intersecting with the side surface; and [0127] identify the location of the side surface by detecting, with the three-dimensional sensor, the three-dimensional coordinate values and the inclination of the wrapping paper that has shifted.

((3)))

[0128] The control system according to ((2))), wherein the processor is configured to: [0129] obtain an attribute value representing an attribute of the sheets wrapped in the wrapping paper; and [0130] setting, by using the obtained attribute value of the sheets, strength of force used by the arm to shift the wrapping paper in the direction intersecting with the side surface of the wrapping paper.

((4)))

[0131] The control system according to ((3))), wherein the obtained attribute value includes a plurality of attribute values representing attributes of the sheets, and the attributes include attributes regarding the number of sheets in the wrapping paper, a size of the sheets, and a ream weight or a basis weight of the sheets.

((5)))

[0132] The control system according to ((4))), wherein the attributes also include an attribute regarding a material of the wrapping paper.

((6)))

[0133] The control system according to one of ((1))) to ((5))), wherein the processor is configured to control the arm so that the cutting device opens the side surface of the wrapping paper at a position included in a range above a center of the side surface and below a top edge of the side surface, in a state in which the wrapping paper is placed on a table so that the image forming surfaces of the sheets wrapped in the wrapping paper oppose a loading surface of the table disposed along a floor.

((7)))

[0134] The control system according to one of ((1))) to ((6))), wherein the processor is configured to control the arm to eliminate a piece of the opened wrapping paper remaining above the sheets after the side surface of the wrapping paper is opened.

((8)))

[0135] The control system according to one of ((1))) to ((7))), wherein the processor is configured to perform control, before causing the arm to grip the sheet bundle exposed outside the opened wrapping paper, to cause the arm to contact the sheets to arrange positions of the sheets so that positions of the outlines of the individual sheets are contained in a predetermined range when the sheets exposed outside the opened wrapping paper are seen from a position above the image forming surfaces of the sheets.

((9)))

[0136] The control system according to one of ((1))) to ((8))), wherein the three-dimensional sensor is a three-dimensional sensor located at a fixed place so as not to change a relative position between the three-dimensional sensor and the wrapping paper.

((10)))

[0137] A control program causing a computer to execute a process comprising: [0138] detecting, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate

values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; [0139] identifying, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; [0140] controlling an arm to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and [0141] removing the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.

((11))

[0142] A control method comprising: [0143] detecting, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; [0144] identifying, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; [0145] controlling an arm to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and [0146] removing the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.

((12))

[0147] A control device comprising: [0148] a processor configured to: [0149] detect, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; [0150] identify, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; [0151] control an arm to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and [0152] remove the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.

Claims

1. A control system comprising: a processor configured to: detect, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; identify, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; control an arm

to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and remove the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.

2. The control system according to claim 1, wherein the processor is configured to: control the arm, before opening the side surface of the wrapping paper, so that the arm contacts the side surface to shift the wrapping paper in a direction intersecting with the side surface; and identify the location of the side surface by detecting, with the three-dimensional sensor, the three-dimensional coordinate values and the inclination of the wrapping paper that has shifted.

3. The control system according to claim 2, wherein the processor is configured to: obtain an attribute value representing an attribute of the sheets wrapped in the wrapping paper; and setting, by using the obtained attribute value of the sheets, strength of force used by the arm to shift the wrapping paper in the direction intersecting with the side surface of the wrapping paper.

4. The control system according to claim 3, wherein the obtained attribute value includes a plurality of attribute values representing attributes of the sheets, and the attributes include attributes regarding the number of sheets in the wrapping paper, a size of the sheets, and a ream weight or a basis weight of the sheets.

5. The control system according to claim 4, wherein the attributes also include an attribute regarding a material of the wrapping paper.

6. The control system according to claim 1, wherein the processor is configured to control the arm so that the cutting device opens the side surface of the wrapping paper at a position included in a range above a center of the side surface and below a top edge of the side surface, in a state in which the wrapping paper is placed on a table so that the image forming surfaces of the sheets wrapped in the wrapping paper oppose a loading surface of the table disposed along a floor.

7. The control system according to claim 2, wherein the processor is configured to control the arm so that the cutting device opens the side surface of the wrapping paper at a position included in a range above a center of the side surface and below a top edge of the side surface, in a state in which the wrapping paper is placed on a table so that the image forming surfaces of the sheets wrapped in the wrapping paper oppose a loading surface of the table disposed along a floor.

8. The control system according to claim 3, wherein the processor is configured to control the arm so that the cutting device opens the side surface of the wrapping paper at a position included in a range above a center of the side surface and below a top edge of the side surface, in a state in which the wrapping paper is placed on a table so that the image forming surfaces of the sheets wrapped in the wrapping paper oppose a loading surface of the table disposed along a floor.

9. The control system according to claim 4, wherein the processor is configured to control the arm so that the cutting device opens the side surface of the wrapping paper at a position included in a range above a center of the side surface and below a top edge of the side surface, in a state in which the wrapping paper is placed on a table so that the image forming surfaces of the sheets wrapped in the wrapping paper oppose a loading surface of the table disposed along a floor.

10. The control system according to claim 5, wherein the processor is configured to control the arm so that the cutting device opens the side surface of the wrapping paper at a position included in a range above a center of the side surface and below a top edge of the side surface, in a state in which the wrapping paper is placed on a table so that the image forming surfaces of the sheets wrapped in the wrapping paper oppose a loading surface of the table disposed along a floor.

11. The control system according to claim 6, wherein the processor is configured to control the arm to eliminate a piece of the opened wrapping paper remaining above the sheets after the side surface of the wrapping paper is opened.

12. The control system according to claim 7, wherein the processor is configured to control the arm to eliminate a piece of the opened wrapping paper remaining above the sheets after the side surface of the wrapping paper is opened.

- 13.** The control system according to claim 8, wherein the processor is configured to control the arm to eliminate a piece of the opened wrapping paper remaining above the sheets after the side surface of the wrapping paper is opened.
- 14.** The control system according to claim 9, wherein the processor is configured to control the arm to eliminate a piece of the opened wrapping paper remaining above the sheets after the side surface of the wrapping paper is opened.
- 15.** The control system according to claim 10, wherein the processor is configured to control the arm to eliminate a piece of the opened wrapping paper remaining above the sheets after the side surface of the wrapping paper is opened.
- 16.** The control system according to claim 11, wherein the processor is configured to perform control, before causing the arm to grip the sheet bundle exposed outside the opened wrapping paper, to cause the arm to contact the sheets to arrange positions of the sheets so that positions of the outlines of the individual sheets are contained in a predetermined range when the sheets exposed outside the opened wrapping paper are seen from a position above the image forming surfaces of the sheets.
- 17.** The control system according to claim 12, wherein the processor is configured to perform control, before causing the arm to grip the sheet bundle exposed outside the opened wrapping paper, to cause the arm to contact the sheets to arrange positions of the sheets so that positions of the outlines of the individual sheets are contained in a predetermined range when the sheets exposed outside the opened wrapping paper are seen from a position above the image forming surfaces of the sheets.
- 18.** The control system according to claim 1, wherein the three-dimensional sensor is a three-dimensional sensor located at a fixed place so as not to change a relative position between the three-dimensional sensor and the wrapping paper.
- 19.** A control method comprising: detecting, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; identifying, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; controlling an arm to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and removing the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.
- 20.** A non-transitory computer readable medium storing a program causing a computer to execute a process comprising: detecting, by using a three-dimensional sensor that measures a shape of an object, three-dimensional coordinate values and an inclination of wrapping paper wrapping a sheet bundle constituted by sheets, the three-dimensional coordinate values and the inclination representing a position and a shape of the wrapping paper, the inclination of the wrapping paper being an inclination with respect to a reference line; identifying, by using the three-dimensional coordinate values and the inclination of the wrapping paper, a location of a side surface of the wrapping paper, the side surface being a surface of the wrapping paper which intersects with image forming surfaces of the sheets wrapped in the wrapping paper; controlling an arm to be used for gripping an object so that a cutting device attached to the arm moves along outlines of the sheets on the identified side surface of the wrapping paper and opens the side surface of the wrapping paper; and removing the sheet bundle from the opened wrapping paper by causing the arm to grip the sheet bundle exposed outside the opened wrapping paper.
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